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To the Editors, *Machine Learning*

Dear Editors,

I am submitting the manuscript **"Certified Symmetry Discovery: Confidence Sets for Lie-Algebra Generators from Noisy, Limited Data"** for consideration as a research article in *Machine Learning*.

What the paper is about. Symmetry is among the most reliable inductive biases in machine learning, and a growing literature discovers symmetries from data rather than assuming them. The dominant approach is linear-algebraic: a candidate generator is a symmetry exactly when a certain linear operator annihilates it, so the symmetries are that operator's nullspace. With real data the operator is estimated, no singular value is exactly zero, and a symmetry is declared when one of them is judged small. The paper shows that this practice has no error control, and that its logic is inverted: declaring a symmetry because a test failed to reject it controls the probability of *missing* a symmetry, not of *inventing* one. In our experiments a valid level-0.05 nullspace rank test declares a symmetry that does not exist in up to 100% of replications.

The paper recasts symmetry discovery as an equivalence test and gives a procedure, SymCert, that carries it out in closed form. The two ingredients are a scale-free measure of symmetry violation, which makes a tolerance mean the same thing across systems and candidate bases, and the observation that this measure is a ratio of norms of linear images of the fitted model coefficients, so a confidence region propagates through it exactly by two trust-region subproblems. The resulting bound holds simultaneously for every candidate generator, so no multiplicity correction and no sample splitting are needed; we show that splitting, one of the two remedies usually suggested here, is in fact counterproductive.

Why *Machine Learning*. The work sits where the journal's readership sits: it is a methodological and theoretical contribution to a problem that the equivariant-learning and data-driven-modelling communities care about, evaluated empirically rather than only asserted. It also answers, in the specific direction of statistical validity, an open question raised in the concluding section of Otto, Zolman, Kutz and Brunton (JMLR, 2025), whose unified nullspace framework the paper builds on.

What may interest reviewers beyond the method. Three empirical findings run against expectation. Additive noise does not manufacture false symmetries; it biases the measured violation upward and so destroys detection instead. The mechanisms that do manufacture them are incomplete coverage of the state space, for which we give a computable factor that says exactly when this is possible, and above all the fitted model class: sparse regression, the standard tool of data-driven model discovery, deletes the term that breaks a weak symmetry and reports an exact one. We report that our own guarantee does not survive that last failure, and show what restores it.

Reproducibility. Ground truth in every benchmark is exact rather than tolerance-based: the symmetry algebras are computed as exact rational nullspaces. The implementation, all raw results, the figure and table generators, a test suite of thirty checks against independent characterisations, and a script that re-verifies all thirty-one numbers quoted in the running text are public at `https://github.com/Kendiukhov/certified-symmetry-discovery`.

Declarations. This manuscript is original, has not been published previously, and is not under consideration by another journal. I am the sole author. I have no competing interests to declare. The work involves no human participants, no animals and no personal data.

Thank you for considering the manuscript.

Yours sincerely,

Ihor Kendiukhov